NURobotics Robotics Software Course

Week 3: Computer Vision

Intro to Computer Vision

Motivation

- Robots need to act in the worlds
- One of the cheapest and richest sensors is a camera
- Understanding camera images is not easy (for a computer)

What Are These?









What Do Humans See?

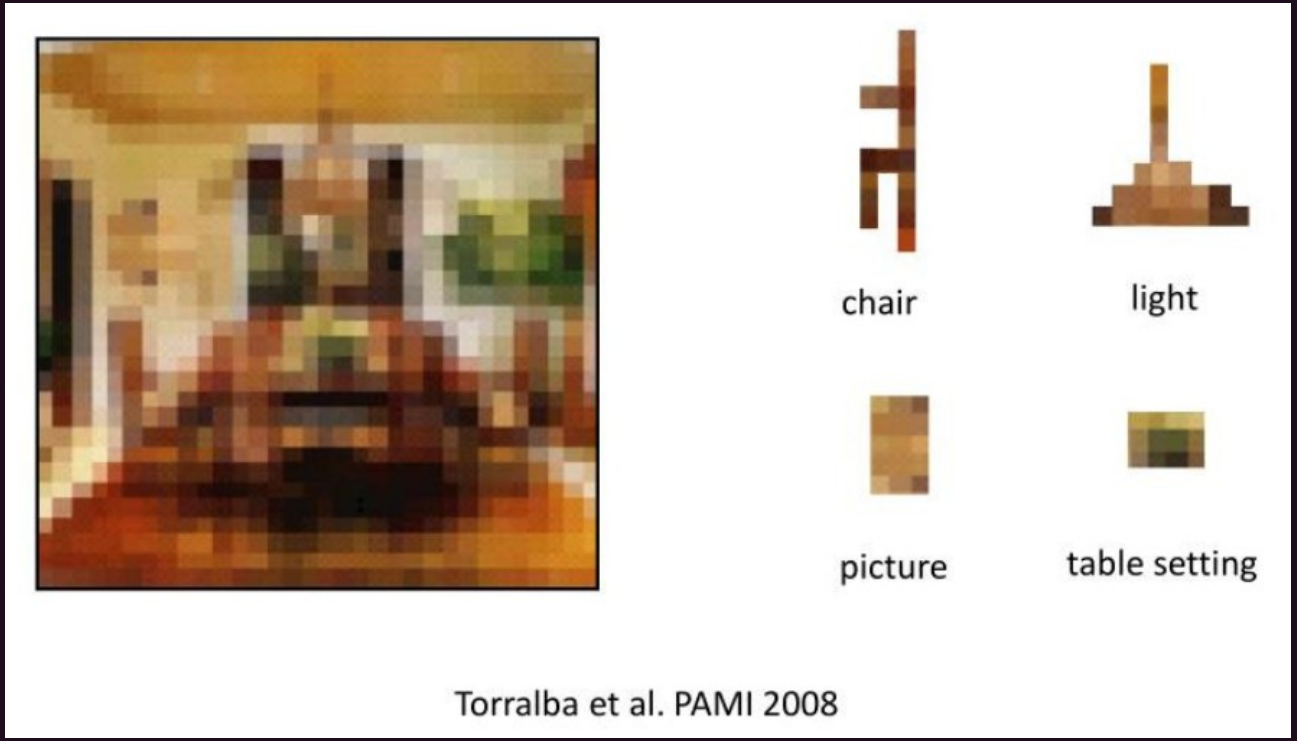
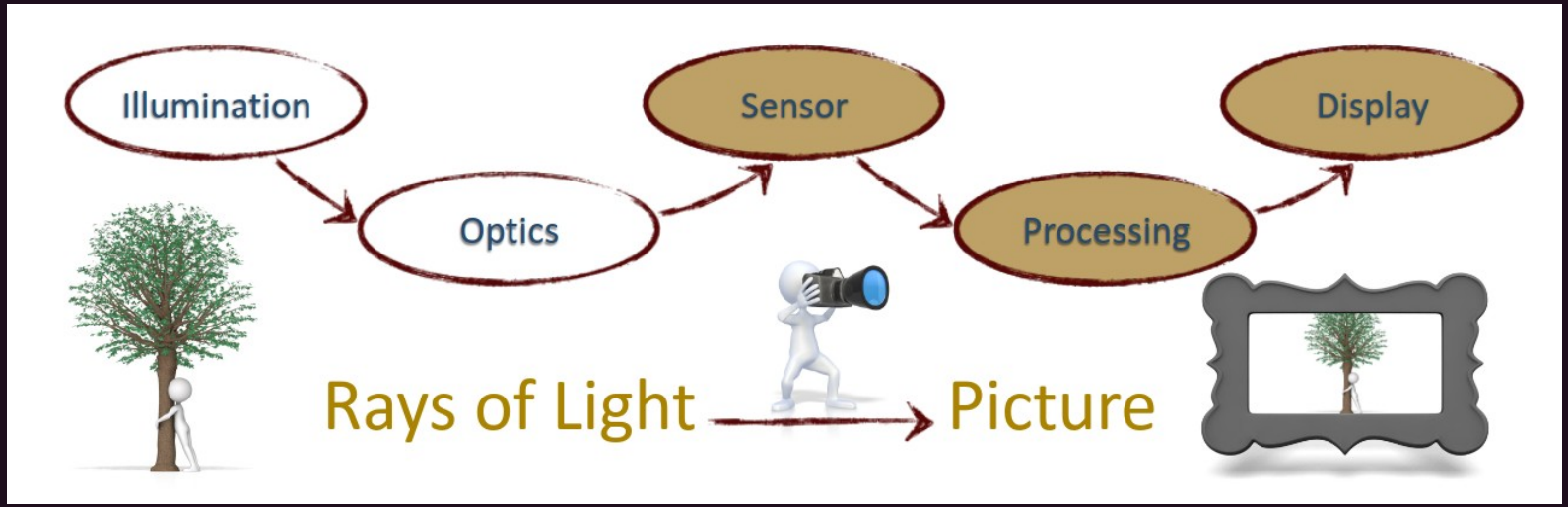
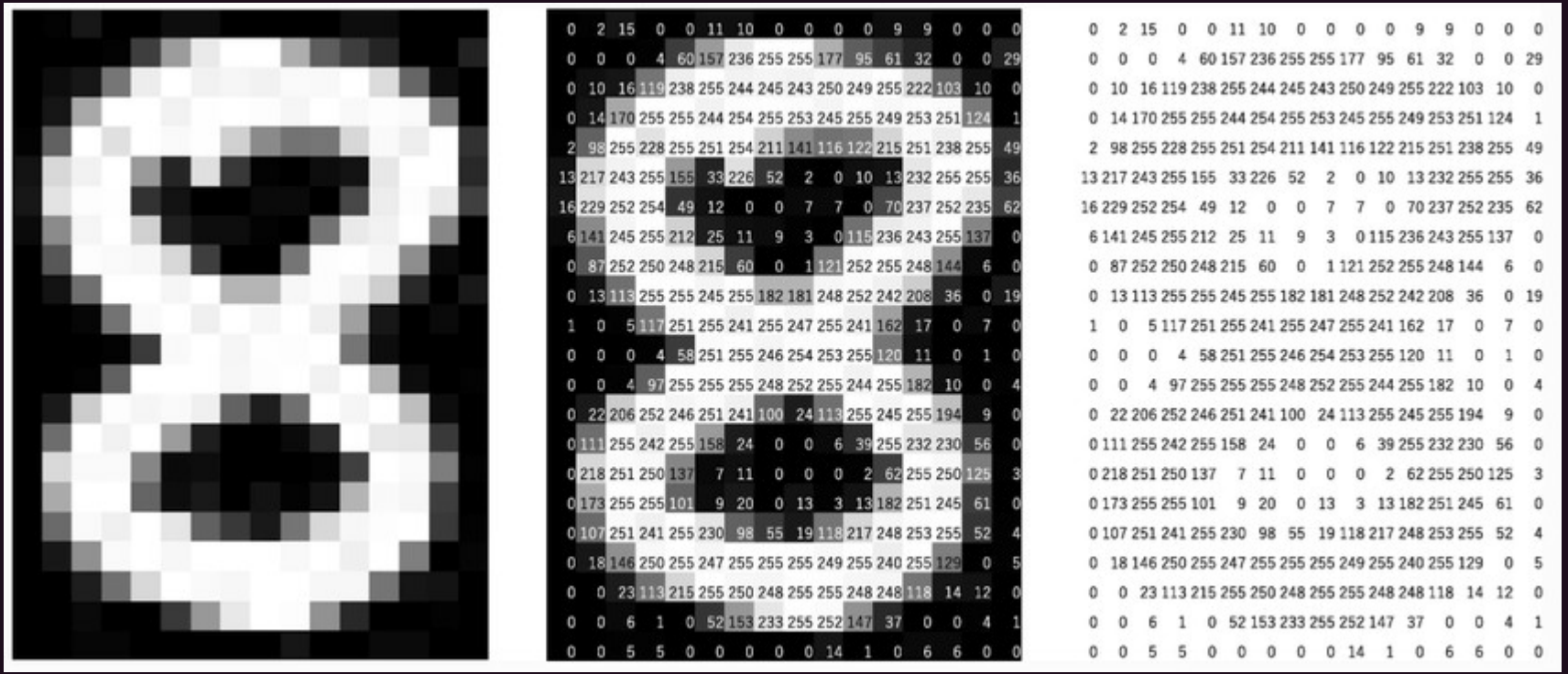


Image Acquisition Pipeline



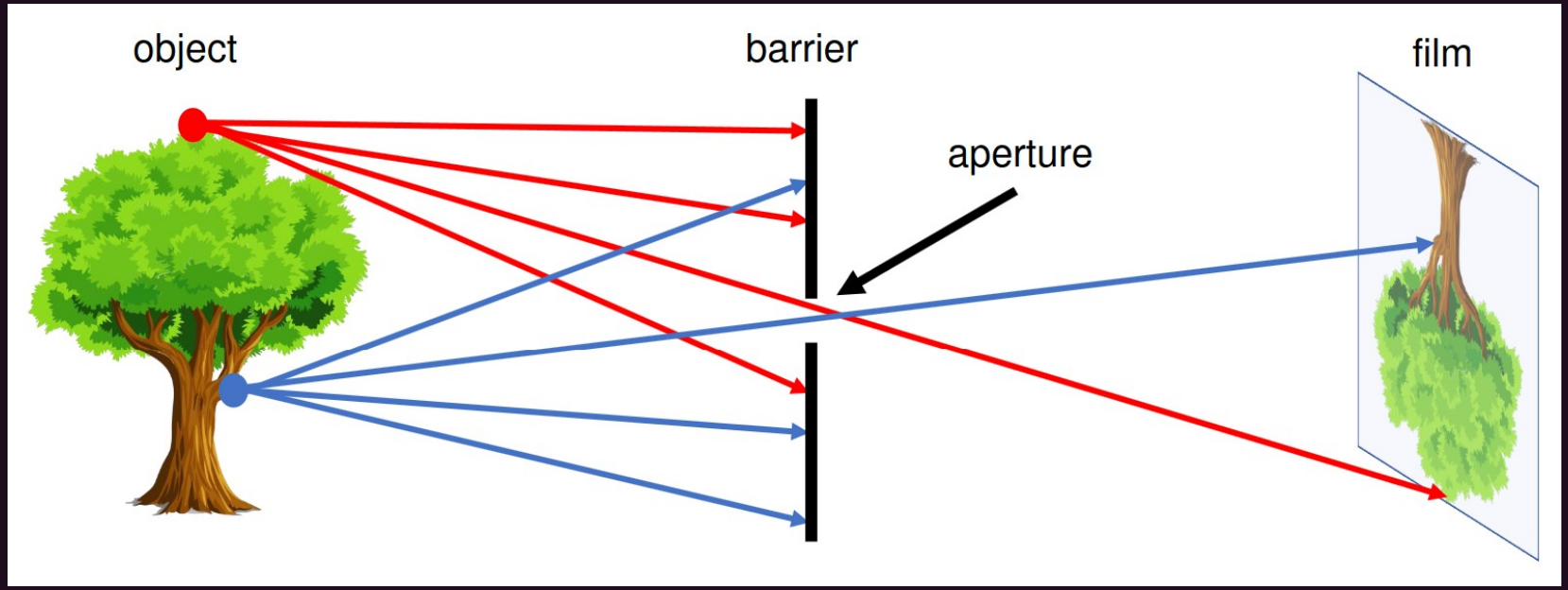
What the Computer Sees



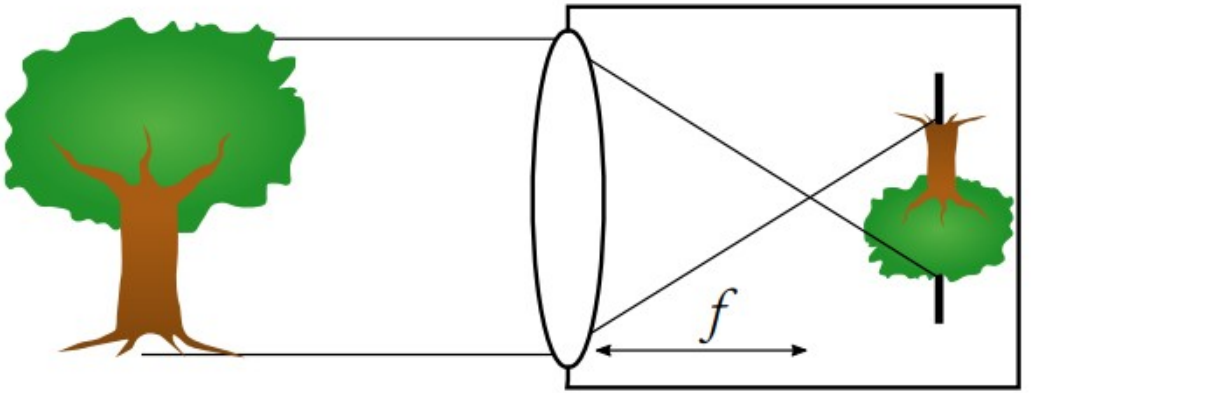
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0	10	16	119	238	255	244	245	243	250	249	255	222	103	10	0
0	14	170	255	255	244	254	255	253	245	255	249	253	251	124	1
2	98	255	228	255	251	254	211	141	116	122	215	251	238	255	49
13	217	243	255	155	33	226	52	2	0	10	13	232	255	255	36
16	229	252	254	49	12	0	0	7	7	0	70	237	252	235	62
6	141	245	255	212	25	11	9	3	0	115	236	243	255	137	0
0	87	252	250	248	215	60	0	1	121	252	255	248	144	6	0
0	13	113	255	255	245	255	182	181	248	252	242	208	36	0	19
1	0	5	117	251	255	241	255	247	255	241	162	17	0	7	0
0	0	0	4	58	251	255	246	254	253	255	120	11	0	1	0
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0	218	251	250	137	7	11	0	0	0	2	62	255	250	125	3
0	173	255	255	101	9	20	0	13	3	13	182	251	245	61	0
0	107	251	241	255	230	98	55	19	118	217	248	253	255	52	4
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0	0	23	113	215	255	250	248	255	255	248	248	118	14	12	0
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0	0	5	5	0	0	0	0	0	14	1	0	6	6	0	0

0	2	15	0	0	11	10	0	0	0	0	9	9	0	0	0
0	0	0	4	60	157	236	255	255	177	95	61	32	0	0	29
0	10	16	119	238	255	244	245	243	250	249	255	222	103	10	0
0	14	170	255	255	244	254	255	253	245	255	249	253	251	124	1
2	98	255	228	255	251	254	211	141	116	122	215	251	238	255	49
13	217	243	255	155	33	226	52	2	0	10	13	232	255	255	36
16	229	252	254	49	12	0	0	7	7	0	70	237	252	235	62
6	141	245	255	212	25	11	9	3	0	115	236	243	255	137	0
0	87	252	250	248	215	60	0	1	121	252	255	248	144	6	0
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0	0	0	4	58	251	255	246	254	253	255	120	11	0	1	0
0	0	4	97	255	255	255	248	252	255	244	255	182	10	0	4
0	22	206	252	246	251	241	100	24	113	255	245	255	194	9	0
0	111	255	242	255	158	24	0	0	6	39	255	232	230	56	0
0	218	251	250	137	7	11	0	0	0	2	62	255	250	125	3
0	173	255	255	101	9	20	0	13	3	13	182	251	245	61	0
0	107	251	241	255	230	98	55	19	118	217	248	253	255	52	4
0	18	146	250	255	247	255	255	255	249	255	240	255	129	0	5
0	0	23	113	215	255	250	248	255	255	248	248	118	14	12	0
0	0	6	1	0	52	153	233	255	252	147	37	0	0	4	1
0	0	5	5	0	0	0	0	0	14	1	0	6	6	0	0

Pinhole Camera



Camera Lens

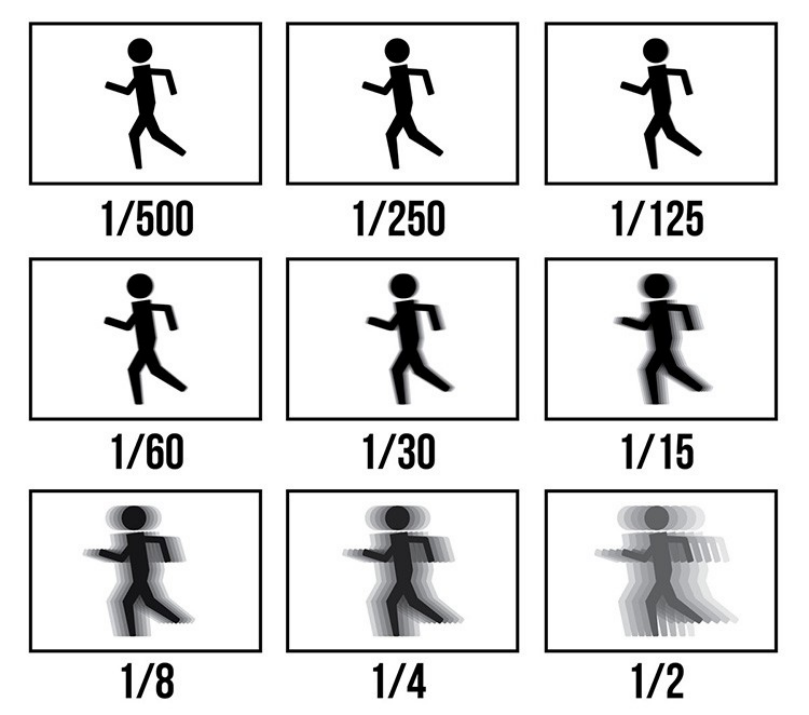


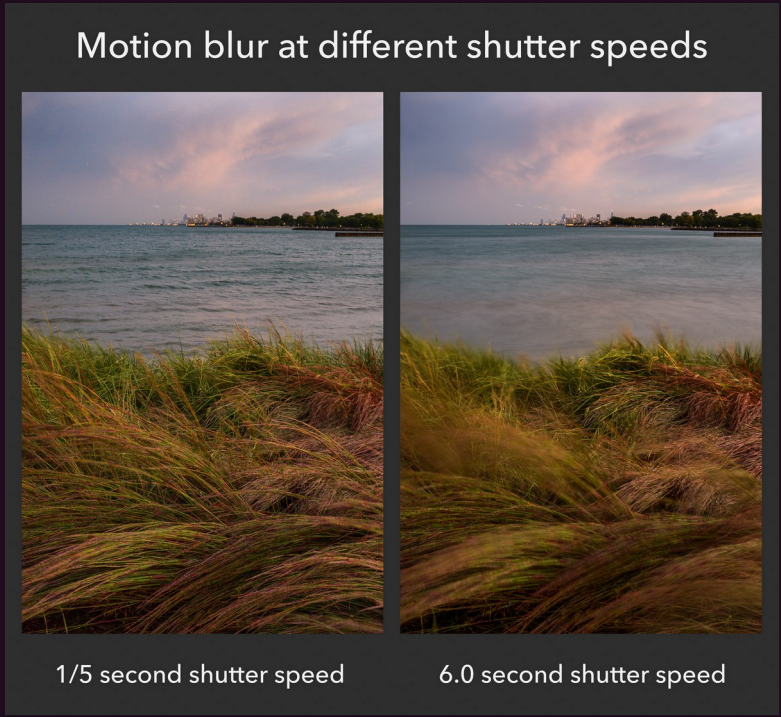
camera with lens to focus image

Camera Properties

- **Resolution**
  - The number of pixels in an image
- **Aperture**
  - Area of light that can enter the lens
- **Field of View (FOV)**
  - Angular measurement that can be seen by the lens
- **Shutter Speed (Exposure Time):**
  - Controls the time to acquire image

Motion Blur

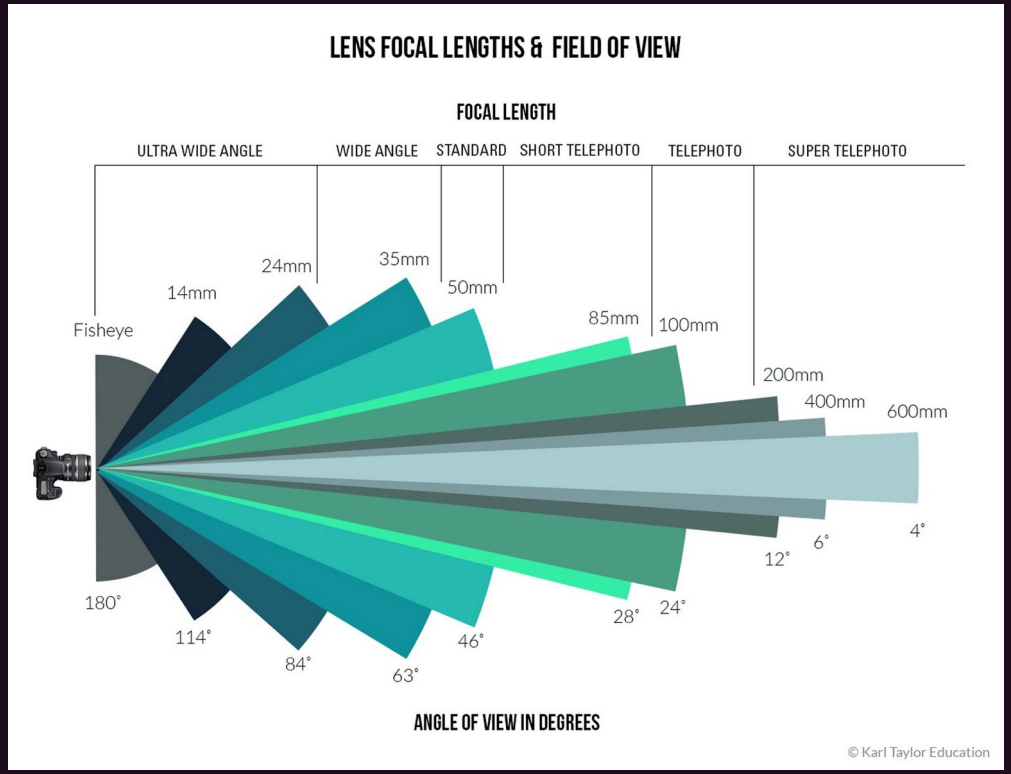

1/500
1/250
1/125
1/60
1/30
1/15
1/8
1/4
1/2


Motion blur at different shutter speeds
1/5 second shutter speed
6.0 second shutter speed

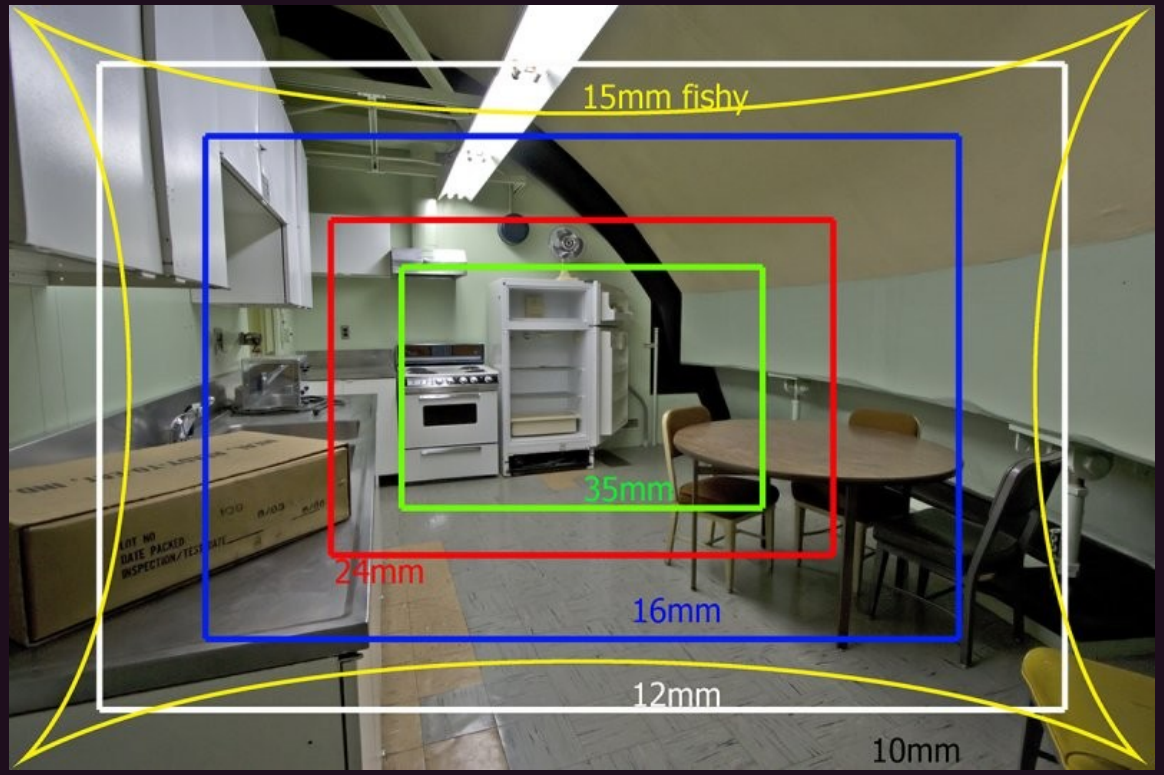
Focal Length



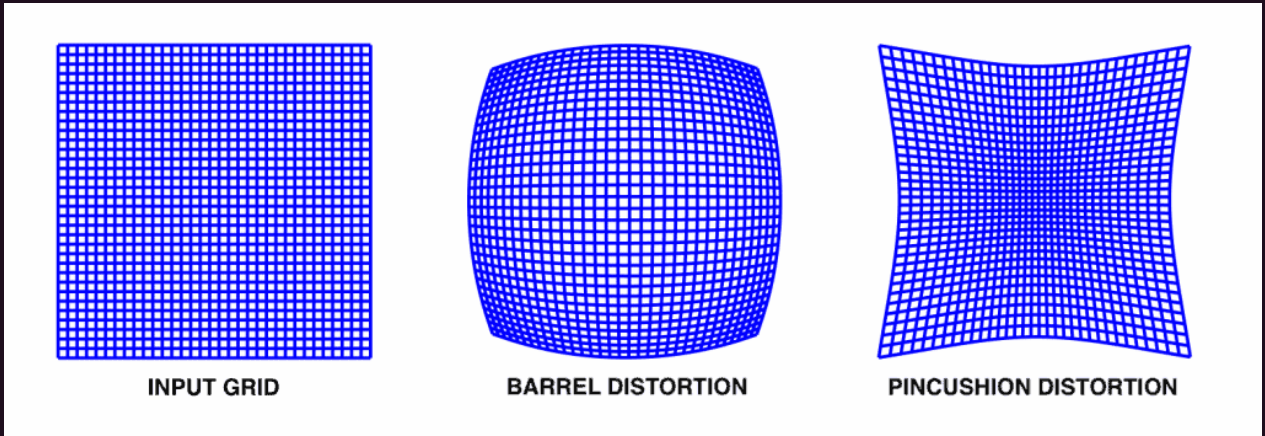
Focal Length

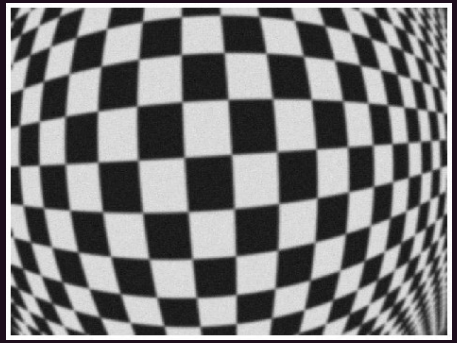


Fisheye Lens

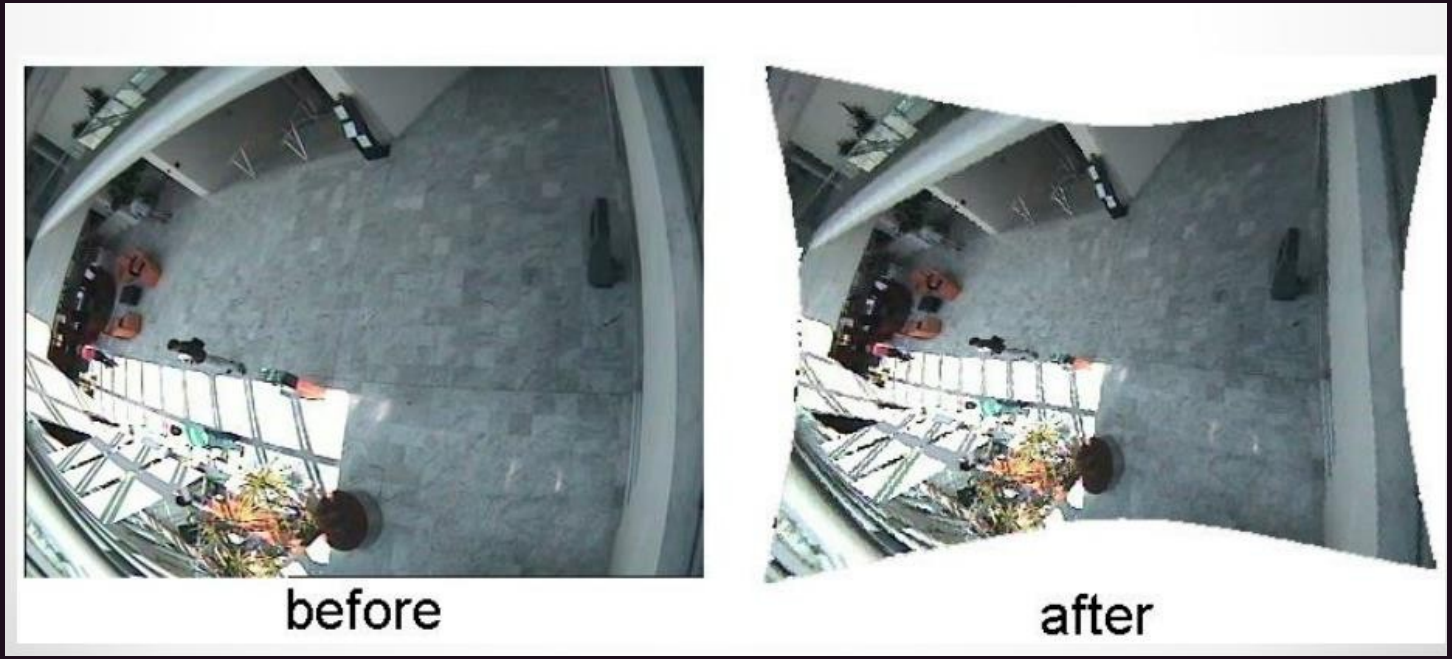


Camera Distortion

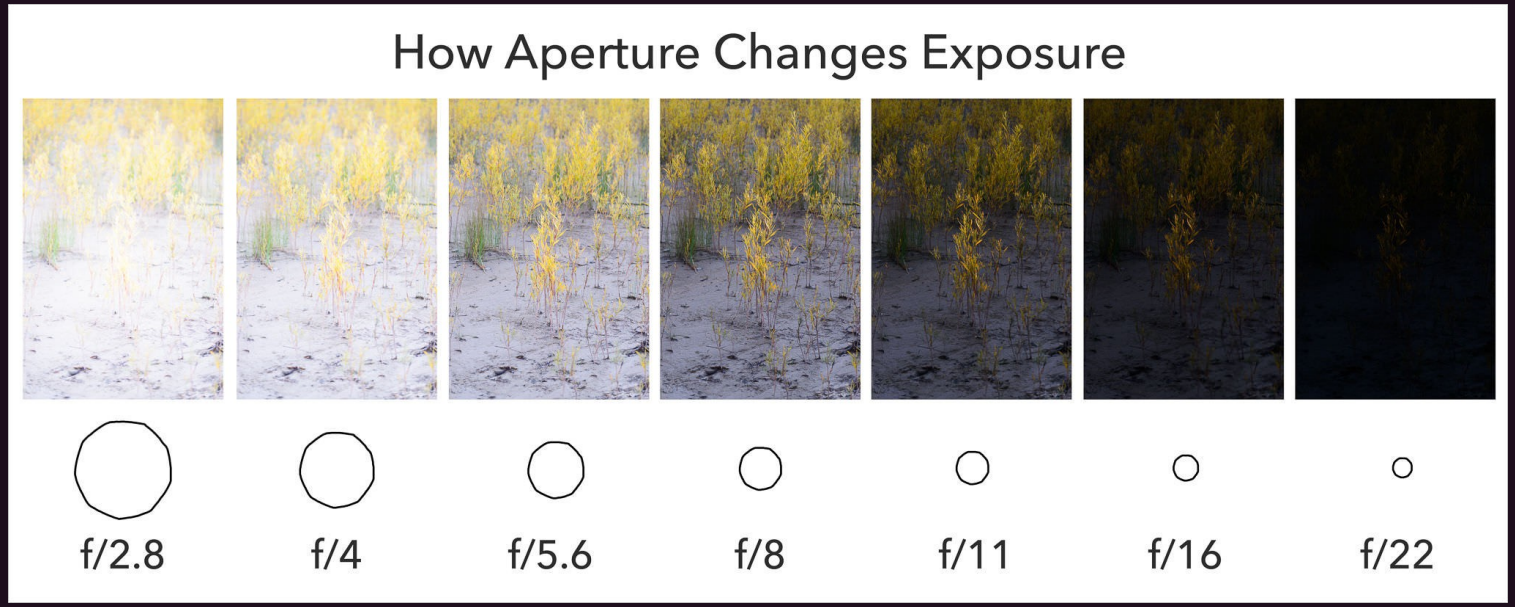




Rectification



Aperture



Color Spaces

- Different encoding for colors
- Five major color models:
  - CIE (first attempt/old)
  - RGB (most common)
  - HSL/HSV (hue/brightness)
  - YUV (luma)
  - CMYK (printing)

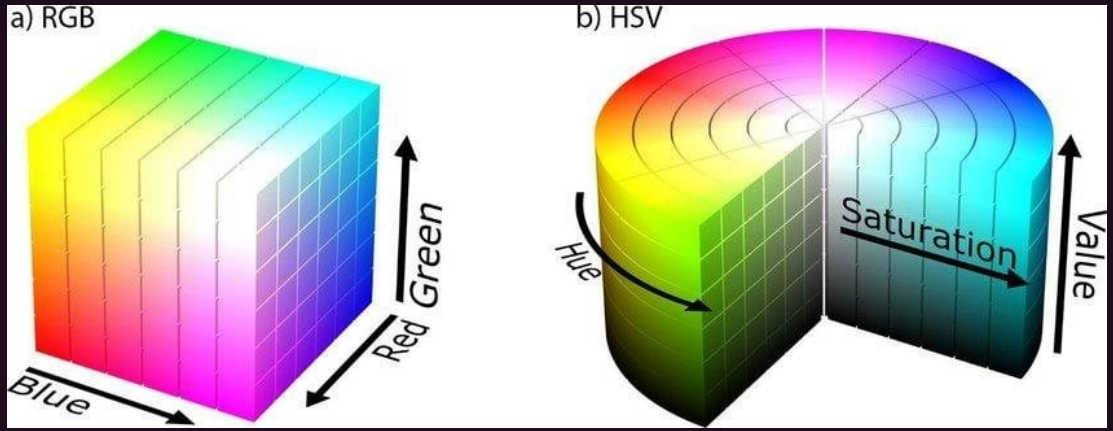
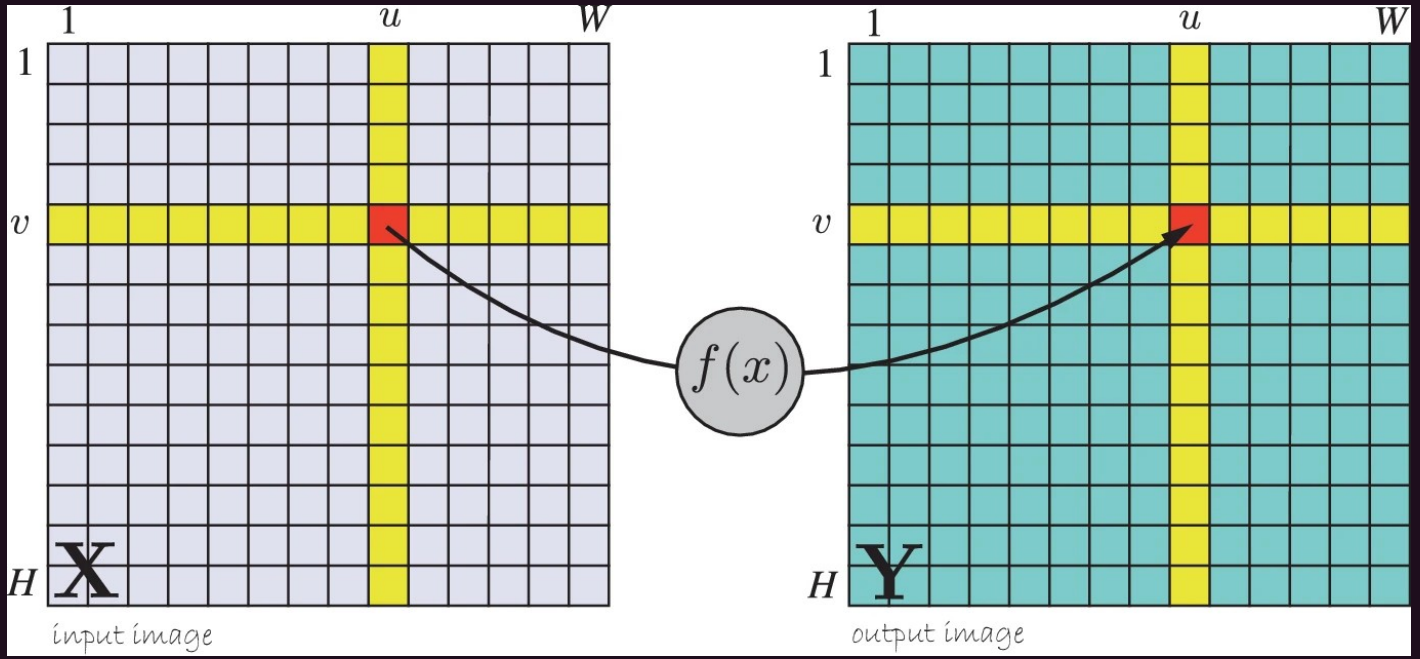


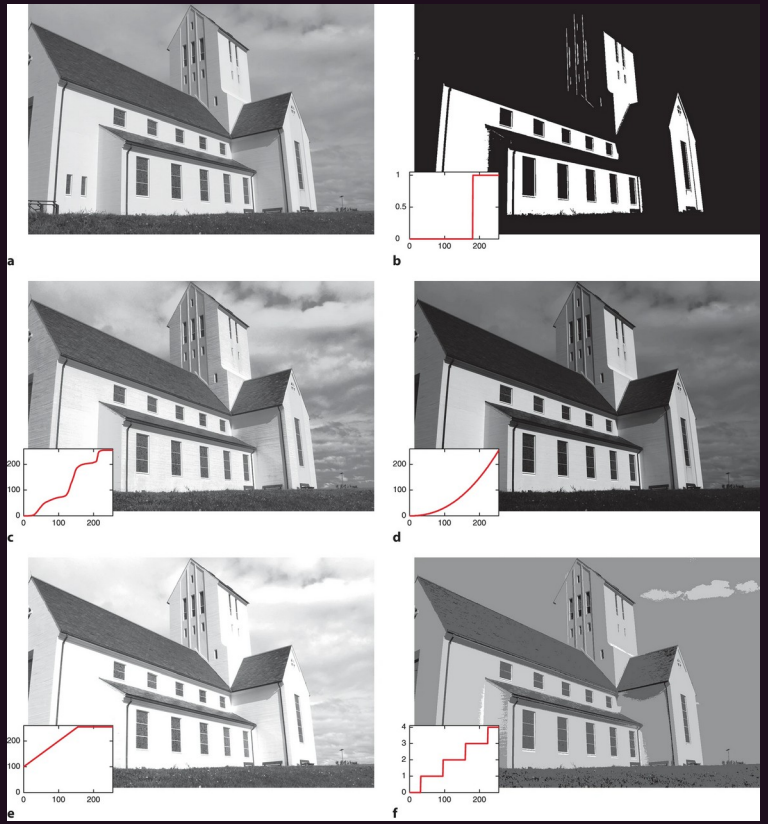
Image Filters

Monadic Operators

Transforms a single pixel input into a single pixel output



Monadic Operator Examples



a. original image

b. thresholded

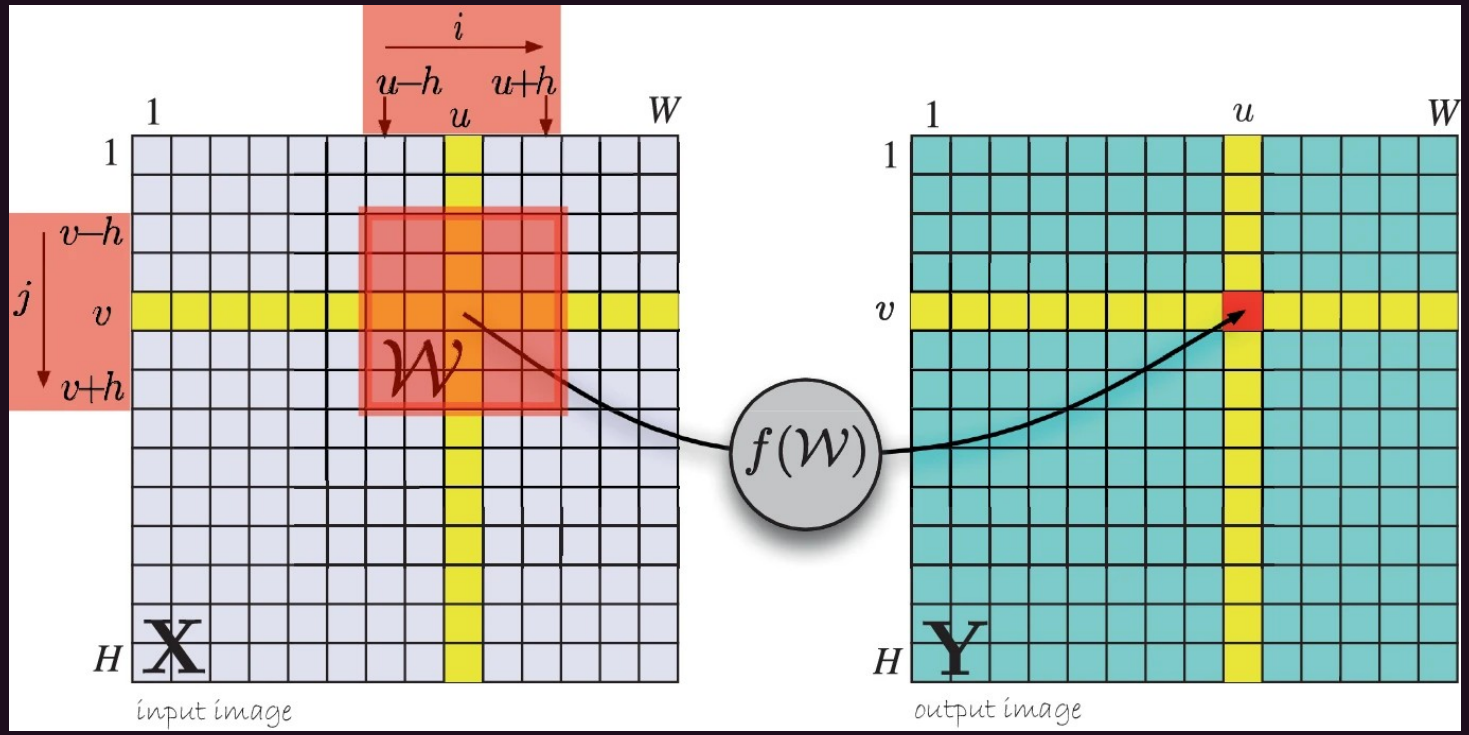
c. histogram normalized

d. gamma corrected

e. after brightness increase

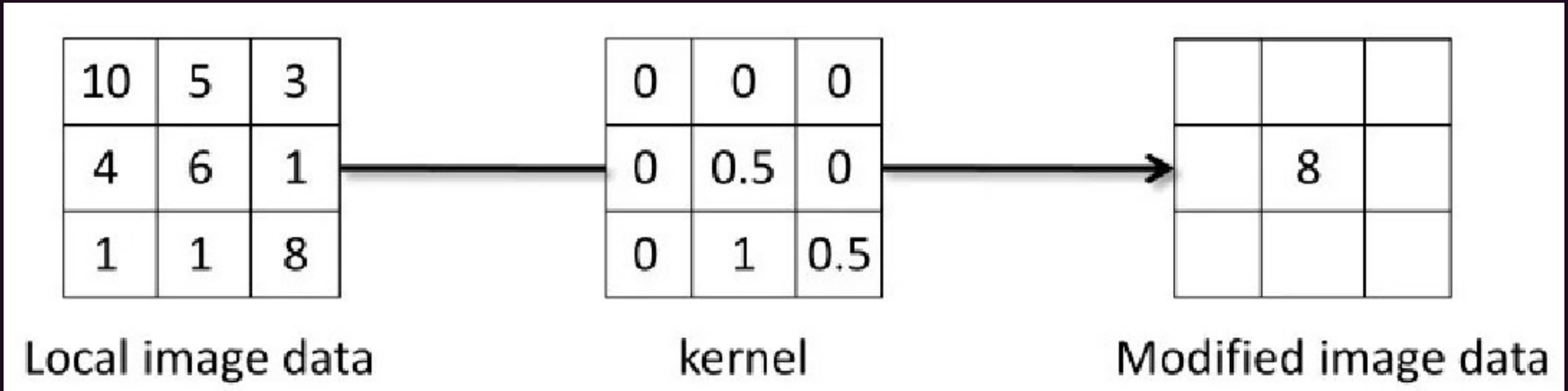
f. posterized

Many-To-One Mapping Between Pixels

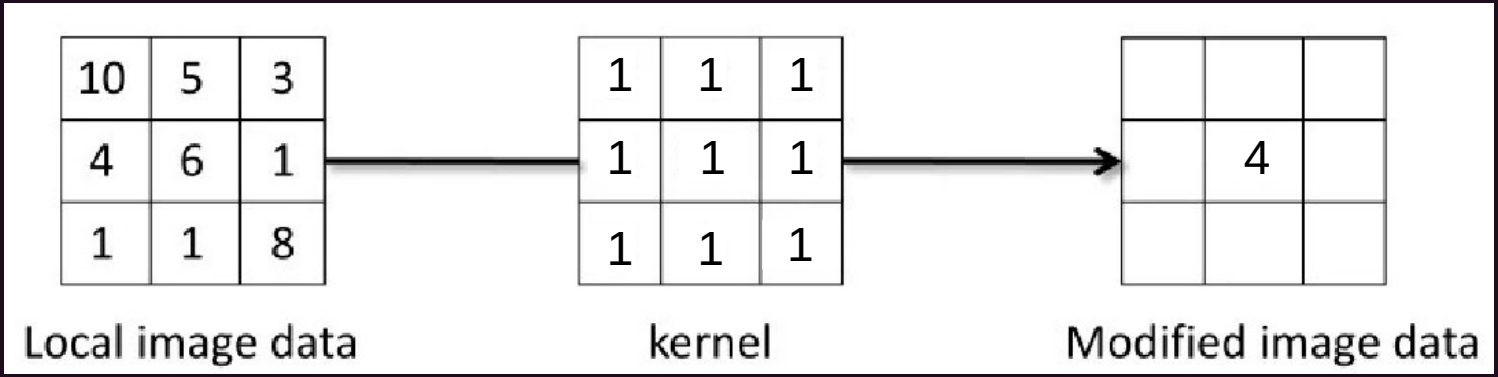


Example: Averaging Using Linear Filtering

- Replace each pixel with a linear combination of its neighbors
- The matrix of the linear combination is called the “kernel,” “mask,” or “filter”



Example: Averaging Using Linear Filtering



Let F be the image, H be the kernel (of size 2k+1 x 2k+1), and G be the output image

$$G[i,j] = \sum_{u=-k}^{k} \sum_{v=-k}^{k} H[u,v]F[i+u, j+v]$$

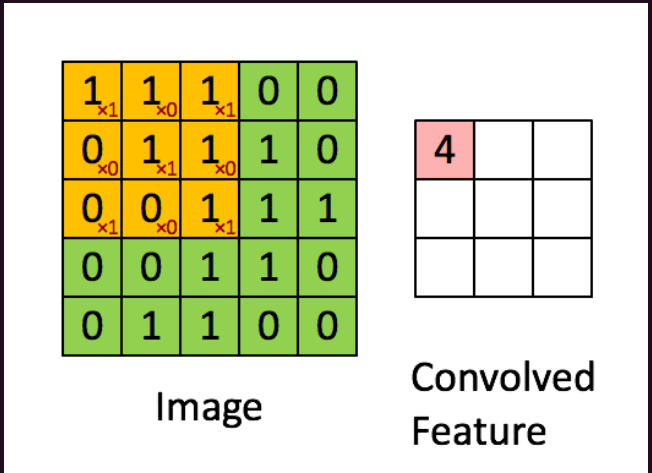
This is called a **cross-correlation** operation:

$$G = H \otimes F$$

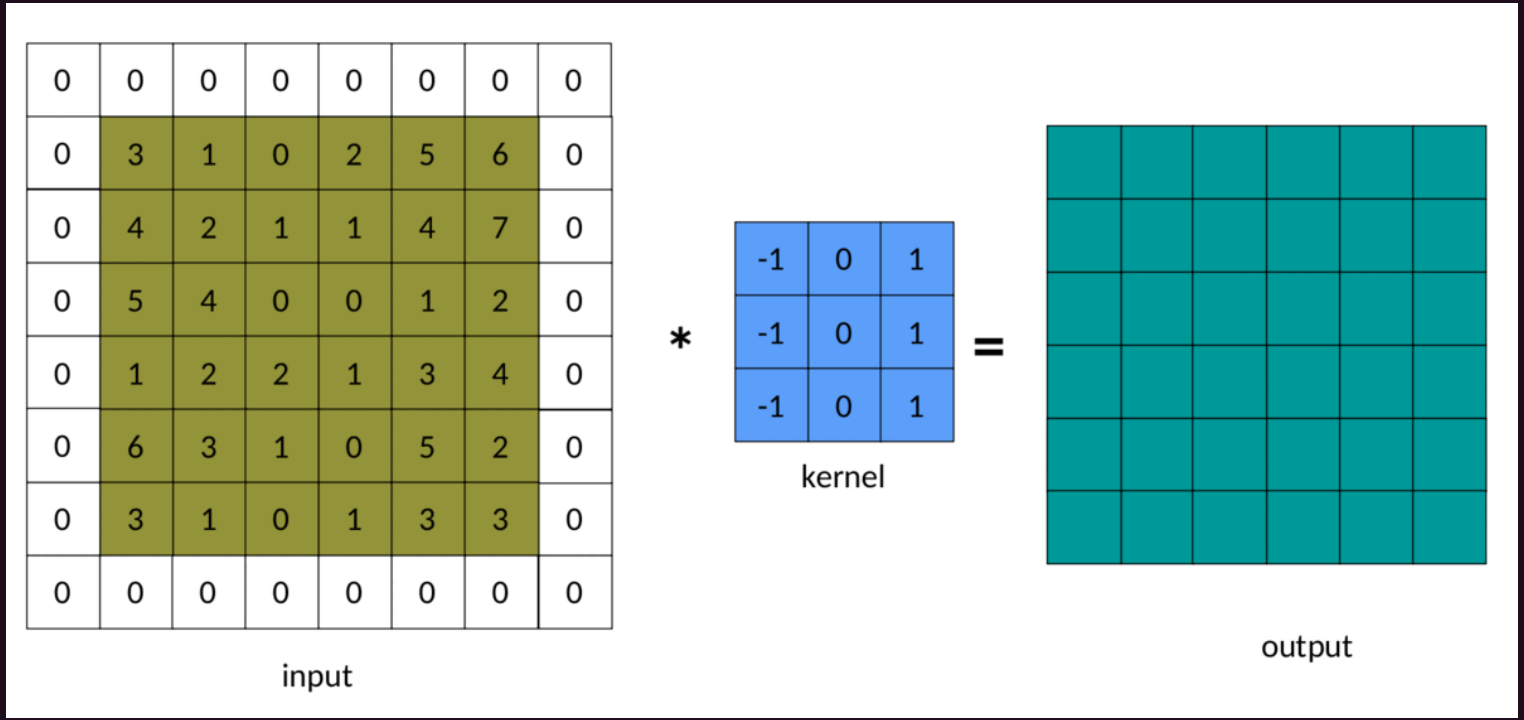
(10*1+5*1+3*1+4*1+6*1+1*1+1*1+1*1+8*1)/9 = 4

Convolution

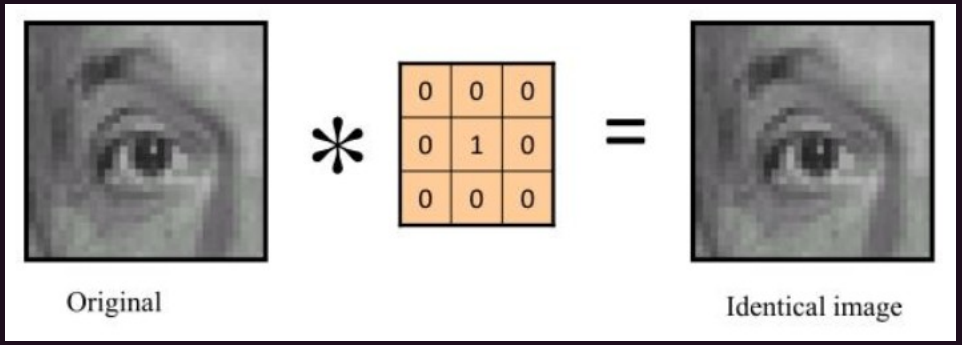
- Applying a kernel to an image in this way is called a convolution
- Convolution can produce unwanted artifacts along the edges of the image (ways to address this include zero padding, edge value replication, mirror extension, and more)
- Stride – The number of rows and columns traversed per slide
- Padding – Added rows/columns needed to perform calculations at image edges

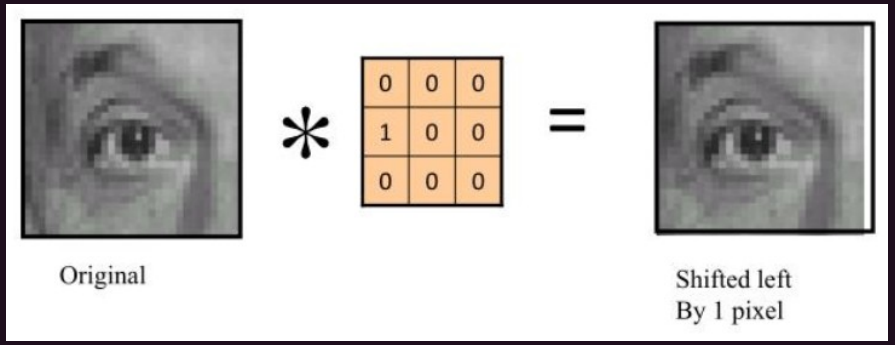


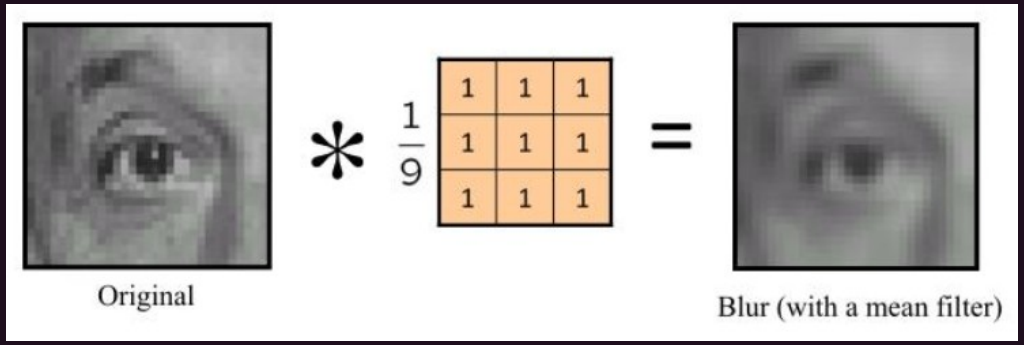
Convolution



Linear Filters Examples

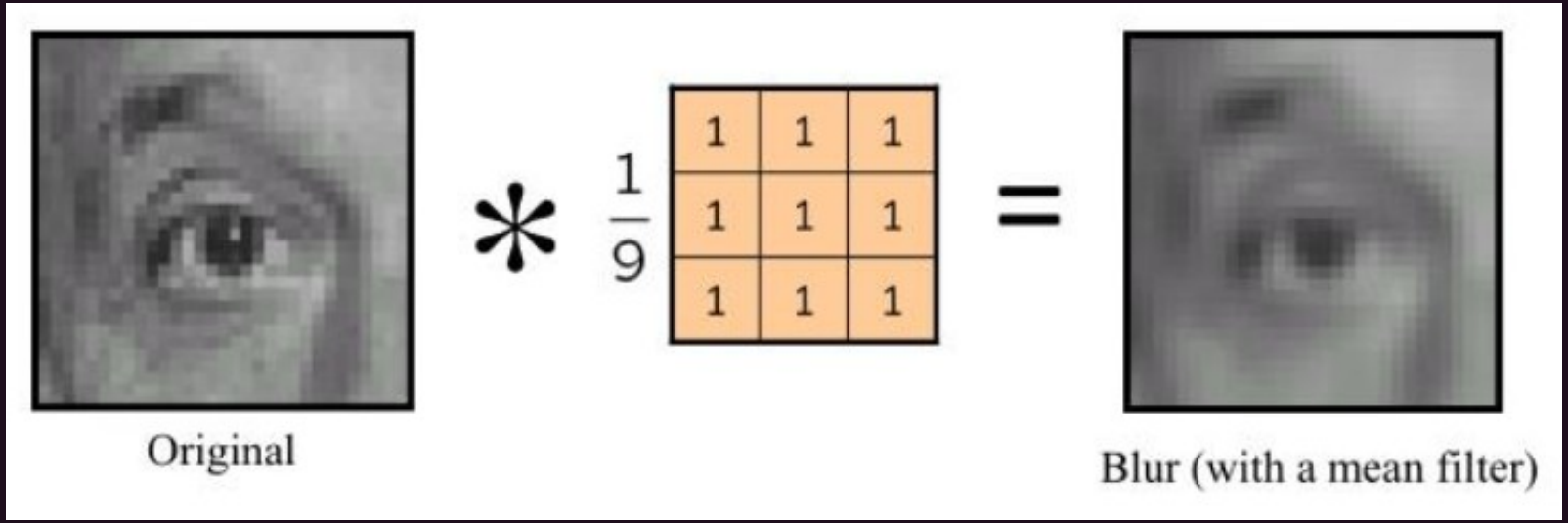




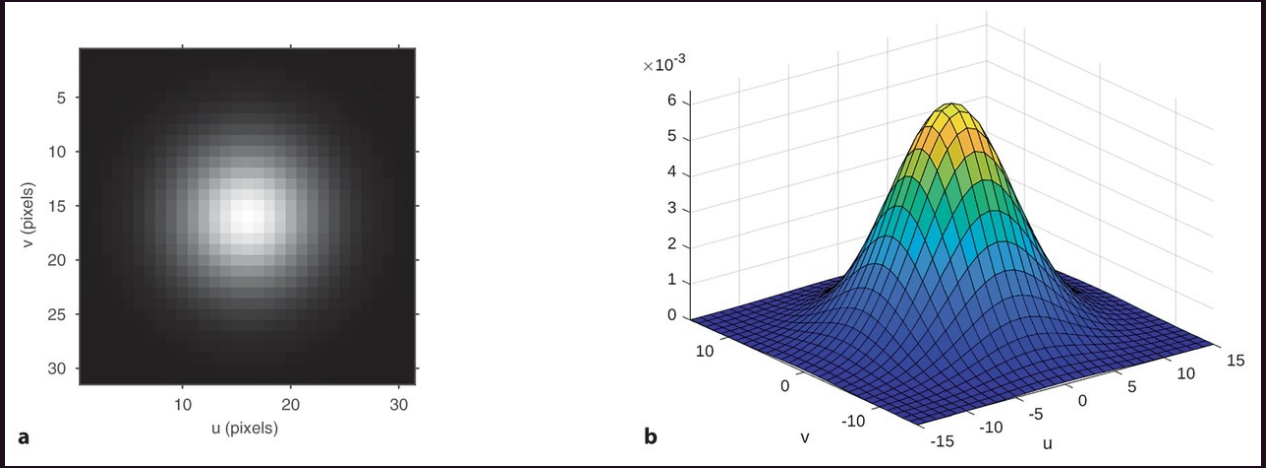


Smoothing Kernels

- Removes noise (high frequency signals)
- All values in the kernel sum to one

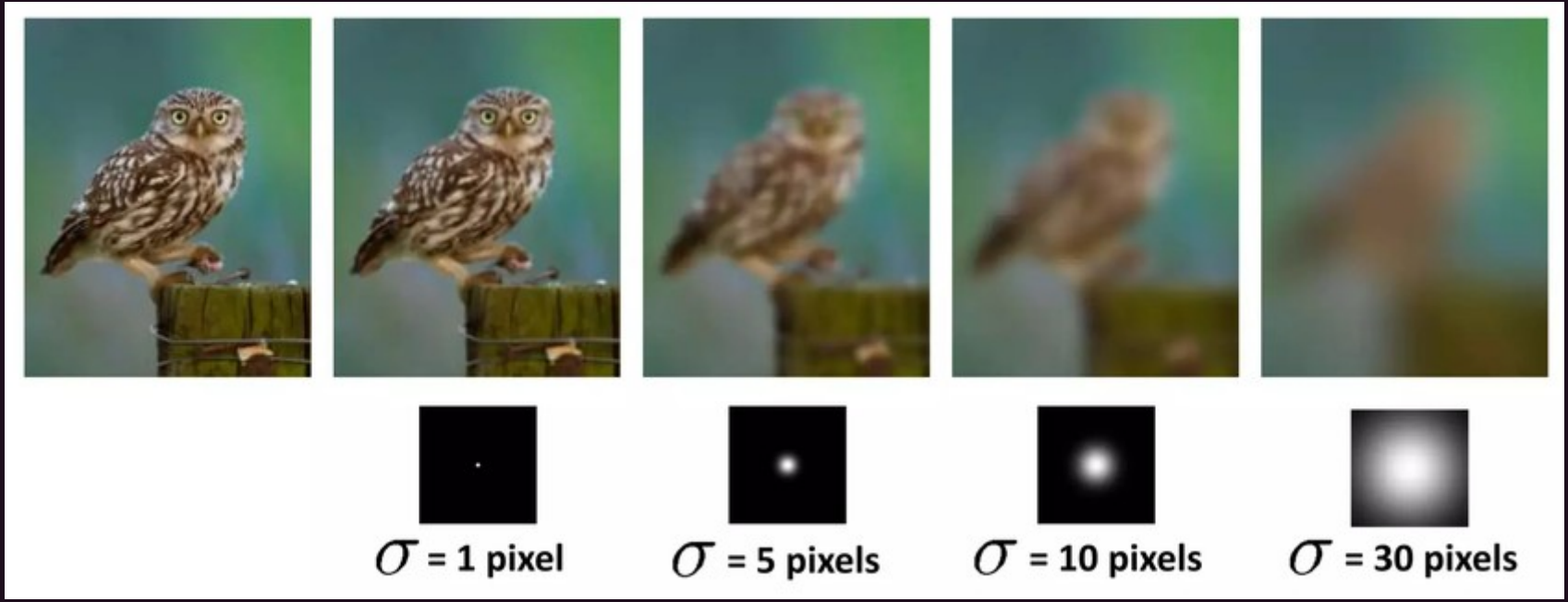


Gaussian Kernel



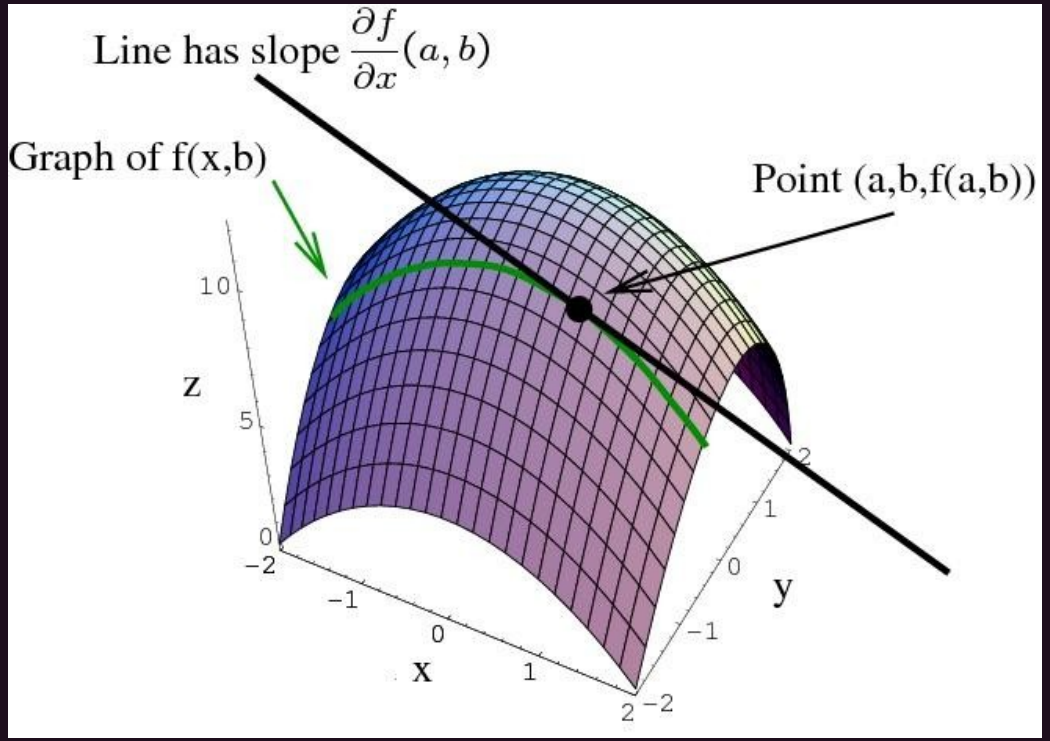
$$\frac{1}{16}\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

Gaussian Filter

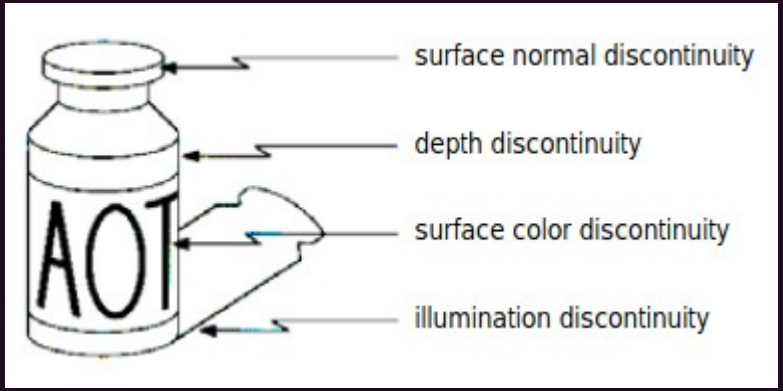


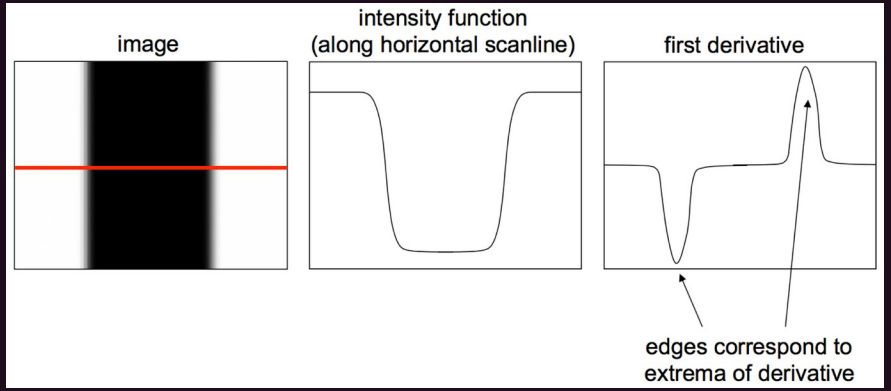
Edge Detection

Multivariable Calculus



Edges


surface normal discontinuity
depth discontinuity
surface color discontinuity
illumination discontinuity
AOT


image
intensity function
(along horizontal scanline)
first derivative
edges correspond to
extrema of derivative

Edge Detection

1	2	1
0	0	0
-1	-2	-1

Sobel Horizontal Mask

-1	0	1
-2	0	2
-1	0	1

Sobel VerticalMask

1	1	1
0	0	0
-1	-1	-1

Prewitt Horizontal Mask

-1	0	1
-1	0	1
-1	0	1

Prewitt Vertical Mask

Sobel and Prewitt Filters (cont)



Original Image

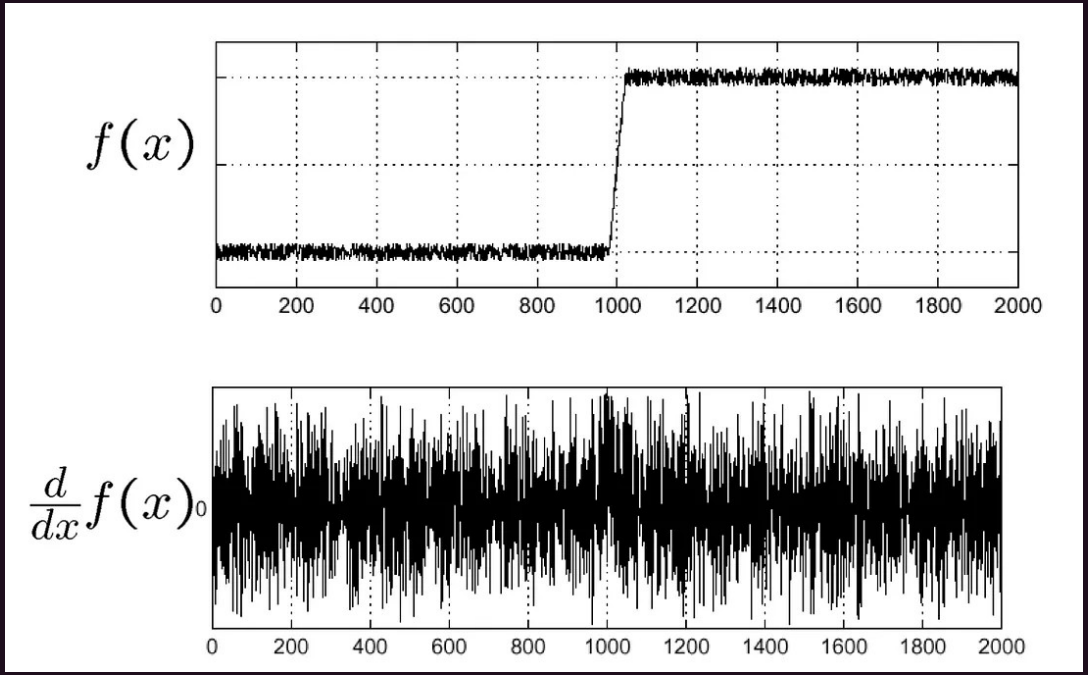


With Sobel Filter Applied

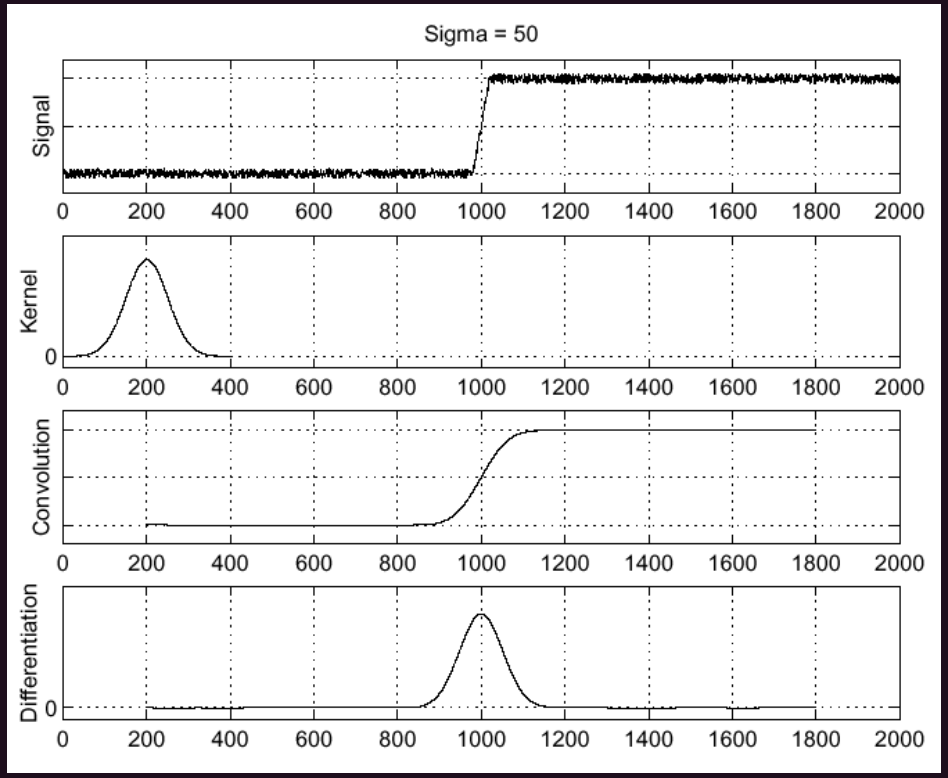


With Prewitt Filter Applied

Why Edge Detections Can Fail



Solution: Smoothing Operation



Smoothing + Edge Detection Results

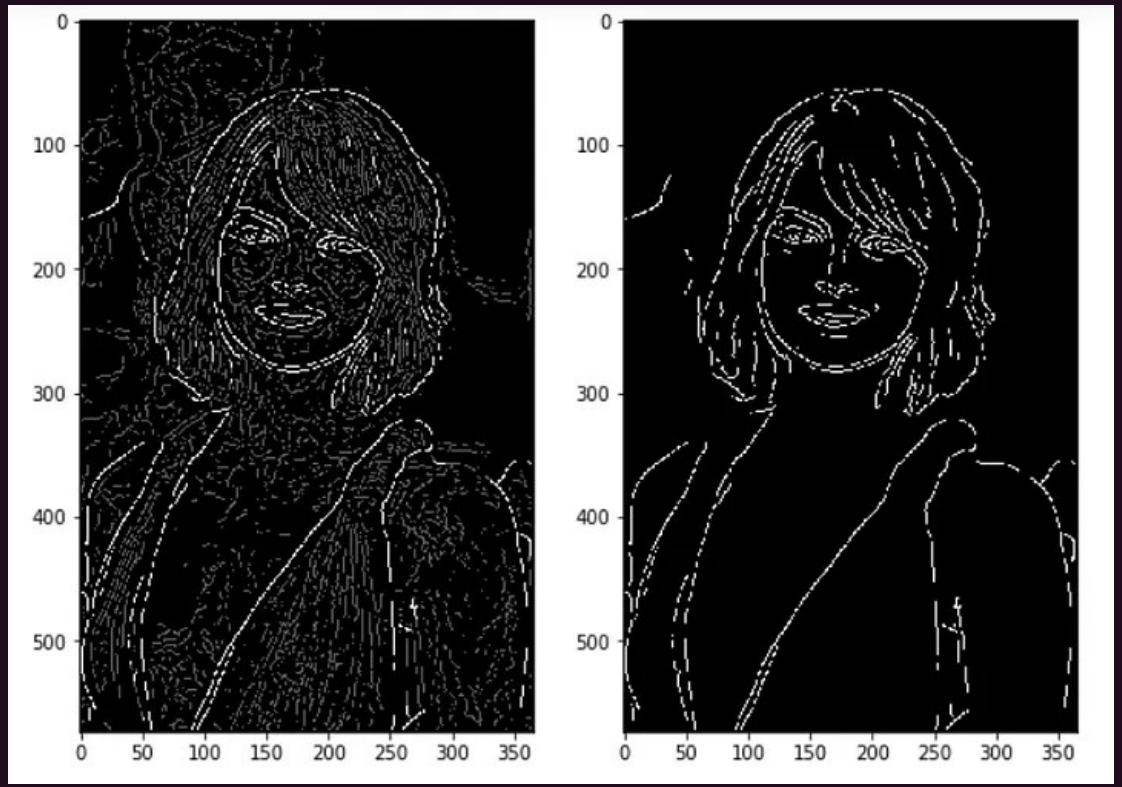


Image Filters

- Good for simple geometric or color-based operations
- Very fast to implement
- Black box magic – tuning is key

Organizing ROS Code